

University of Moratuwa
Department of Electronic and Telecommunication Engineering
EN2091 Laboratory Practice and Projects

Metal Detector

Feasibility Report

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Team Prometheus Members

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Introduction

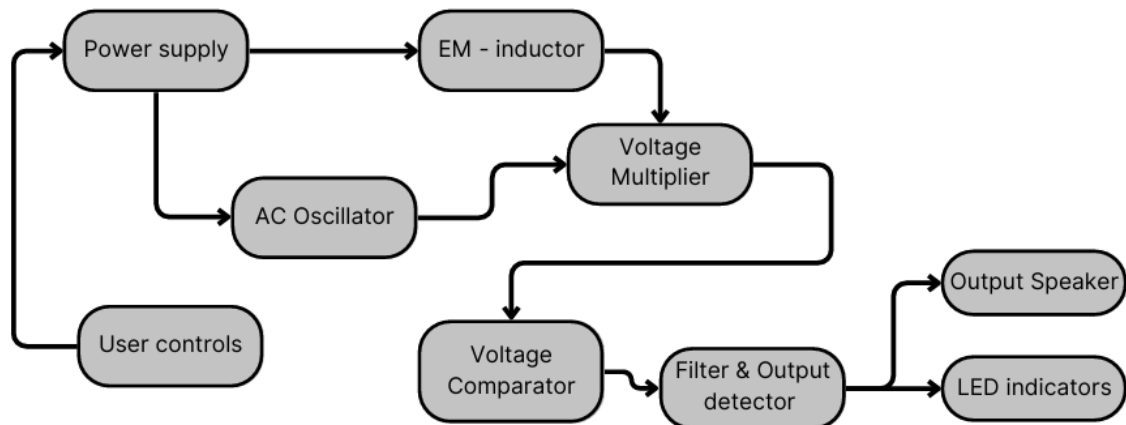
Metal detectors are devices designed to sense the presence of metallic objects by detecting the variations caused by them to an electromagnetic field. These devices find applications in security, archaeology, and industrial sectors. Our project aims to implement a hand-held metal detector, most commonly seen in security applications using transistors, operational amplifiers and RCL components.

Functionality

We are aiming to create a handheld metal detector device with following specifications,

- Detect metallic objects at distances up to 10 cm.
- Provide an audible alert and a visual alert when a metallic object is detected.
- Indicate battery level and operation using LEDs.
- Operate using a 9V replaceable battery for portability.
- Be compact and lightweight for easy handling.
- Operate at 25KHz frequency.

Functional Block Diagram



Methodology

The proposed metal detector operates by using an oscillator circuit to generate a stable signal, which changes in frequency when metal objects are nearby due to inductance variations. This signal is processed through a voltage multiplier to amplify the effect, and a comparator circuit is used to indicate the detection of presence of metal.

Micro-products

- Schmitt Trigger Oscillator.
- Power supply and Voltage regulator.
- Electro-Magnetic inductor.
- Voltage multiplier and RC filter.
- Voltage comparators.
- High pass filter.
- Output indicators and speaker.

Micro-products Allocation

Team Member	Allocated Micro-product(s)
Eranga W.A.O. 230175U	Electro-magnetic inductor.
Gamage S.K. 230195F	Schmitt trigger oscillator.
Samarasinghe S.M.R.R. 230566U	Voltage multiplier and RC filter.
Tharushika G.K.E. 230636K	Voltage comparator, High pass filter and output indicators.

References

1. ElectroBOOM on analog metal detectors: <https://www.youtube.com/watch?v=yzrVHEB02Hw>
2. Metal Detector Circuit: <https://journalspub.com/publication/ijaic/article=16354/>
3. Model of a Novel High-Sensitivity Metal Detector Coil: <https://ieeexplore.ieee.org/document/10884727>